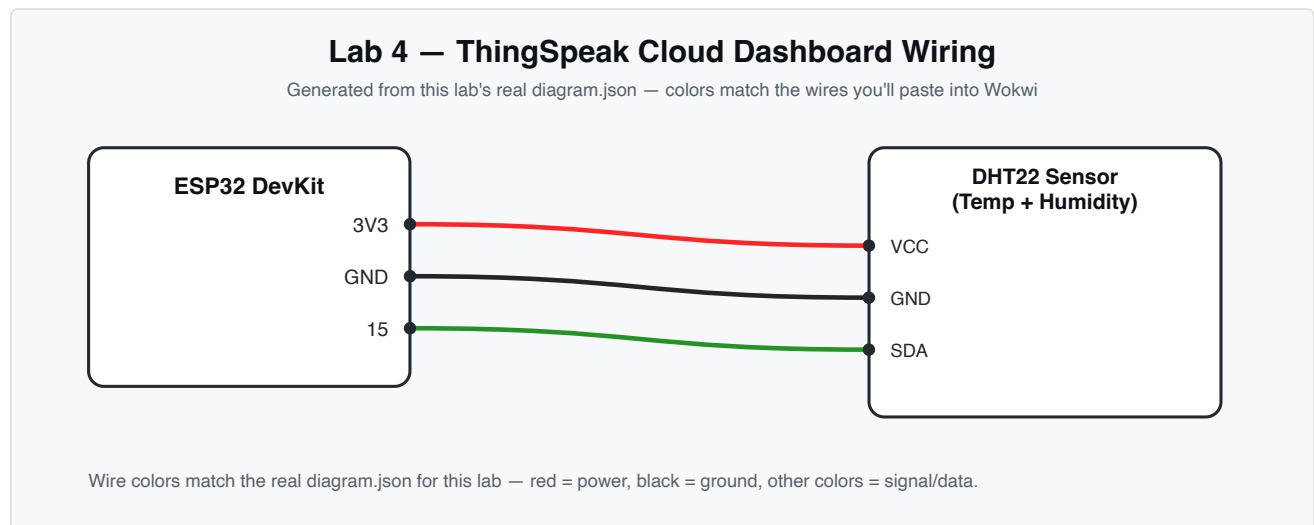


Lab 4 — Cloud Dashboard with ThingSpeak

Time: ~30 minutes **You'll learn:** how to send sensor data to a cloud platform over HTTP, and get a live graph without writing any dashboard/website code yourself.

Wiring diagram



ESP32 Pin	Wire Color	Connects To
3V3	red	DHT22 Sensor (VCC)
GND	black	DHT22 Sensor (GND)
15	green	DHT22 Sensor (data pin)

Same wiring as Labs 2 and 3 — this lab is entirely about what the code does with the reading once it has one.

Part A — Create your ThingSpeak channel (5 min)

1. Go to thingspeak.com and click **Sign In**. Use the free MathWorks account you made in the [Getting Started guide](#).

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To use ThingSpeak, you must sign in with your existing MathWorks account or create a new one.

Non-commercial users may use ThingSpeak for free. Free accounts offer limits on certain functionality. Commercial users are eligible for a time-limited free evaluation. To get full access to the MATLAB analysis features on ThingSpeak, log in to ThingSpeak using the email address associated with your university or organization.

To send data faster to ThingSpeak or to send more data from more devices, consider the [paid license options](#) for commercial, academic, home and student usage.

Sign In

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2. Click **Channels** → **My Channels**.

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My Channels

New Channel

My Channels

My Image Channels

Watched Channels

Public Channels

search by tag

Q

Name	Created	Updated
My IoT Workshop Station Private Public Settings Sharing API Keys Data Import / Export	2026-10-01	2026-10-01 05:32

Help

Collect data in a ThingSpeak channel from a device, from another channel, or from the web.

Click New Channel to create a new ThingSpeak channel.

Click on the column headers of the table to sort by the entries in that column or click on a tag to show channels with that tag.

Learn to [create channels](#), explore and transform data.

Learn more about [ThingSpeak Channels](#).

Examples

- [Arduino](#)
- [Arduino MKR1000](#)
- [ESP8266](#)
- [Raspberry Pi](#)

Upgrade

Need to send more data faster?

Need to use ThingSpeak for a commercial project?

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3. Click **New Channel**.

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My Channels

New Channel

Search by tag

Q

Name	Created	Updated
My IoT Workshop Station Private Public Settings Sharing API Keys Data Import / Export	2026-10-01	2026-10-01 05:32

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4. Fill in the form: - **Name:** for example "My IoT Workshop Station". - Tick **Field 1** and name it **Temperature** . - Tick **Field 2** and name it **Humidity** .

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New Channel

Name

My IoT Workshop Station

Description

Temperature and humidity from the Lab 4 simulated ESP32 + DHT22

Field 1

Temperature

☒

Field 2

Humidity

☒

Field 3

☐

Field 4

☐

Field 5

☐

Field 6

☐

Field 7

☐

Field 8

☐

Metadata

Tags

(Tags are comma separated)

Help

Channels store all the data that a ThingSpeak application collects. Each channel includes eight fields that can hold any type of data, plus three fields for location data and one for status data. Once you collect data in a channel, you can use ThingSpeak apps to analyze and visualize it.

Channel Settings

- **Percentage complete:** Calculated based on data entered into the various fields of a channel. Enter the name, description, location, URL, video, and tags to complete your channel.
- **Channel Name:** Enter a unique name for the ThingSpeak channel.
- **Description:** Enter a description of the ThingSpeak channel.
- **Field#:** Check the box to enable the field, and enter a field name. Each ThingSpeak channel can have up to 8 fields.
- **Metadata:** Enter information about channel data, including JSON, XML, or CSV data.
- **Tags:** Enter keywords that identify the channel. Separate tags with commas.
- **Link to External Site:** If you have a website that contains information about your ThingSpeak channel, specify the URL.
- **Show Channel Location:**
 - **Latitude:** Specify the latitude position in decimal degrees. For example, the latitude of the city of London is 51.5072.
 - **Longitude:** Specify the longitude position in decimal degrees. For example, the longitude of the city of London is -0.1275.
 - **Elevation:** Specify the elevation position meters. For example, the elevation of the city of London is 35.052.
- **Video URL:** If you have a YouTube™ or Vimeo® video that displays your channel information, specify the full path of the video URL.
- **Link to GitHub:** If you store your ThingSpeak code on GitHub®, specify the GitHub repository URL.

5. Scroll to the bottom of the form and click **Save Channel**.

ThingSpeak™ Channels Apps Devices Support Commercial Use How to Buy ML

Metadata

Tags

(Tags are comma separated)

Link to External Site

Link to GitHub

Elevation

Show Channel Location ☐

Latitude

Longitude

Show Video ☐

☒ YouTube

☐ Vimeo

Video URL

Show Status ☐

Save Channel

Using the Channel

You can get data into a channel from a device, website, or another ThingsSpeak channel. You can then visualize data and transform it using [ThingSpeak Apps](#).

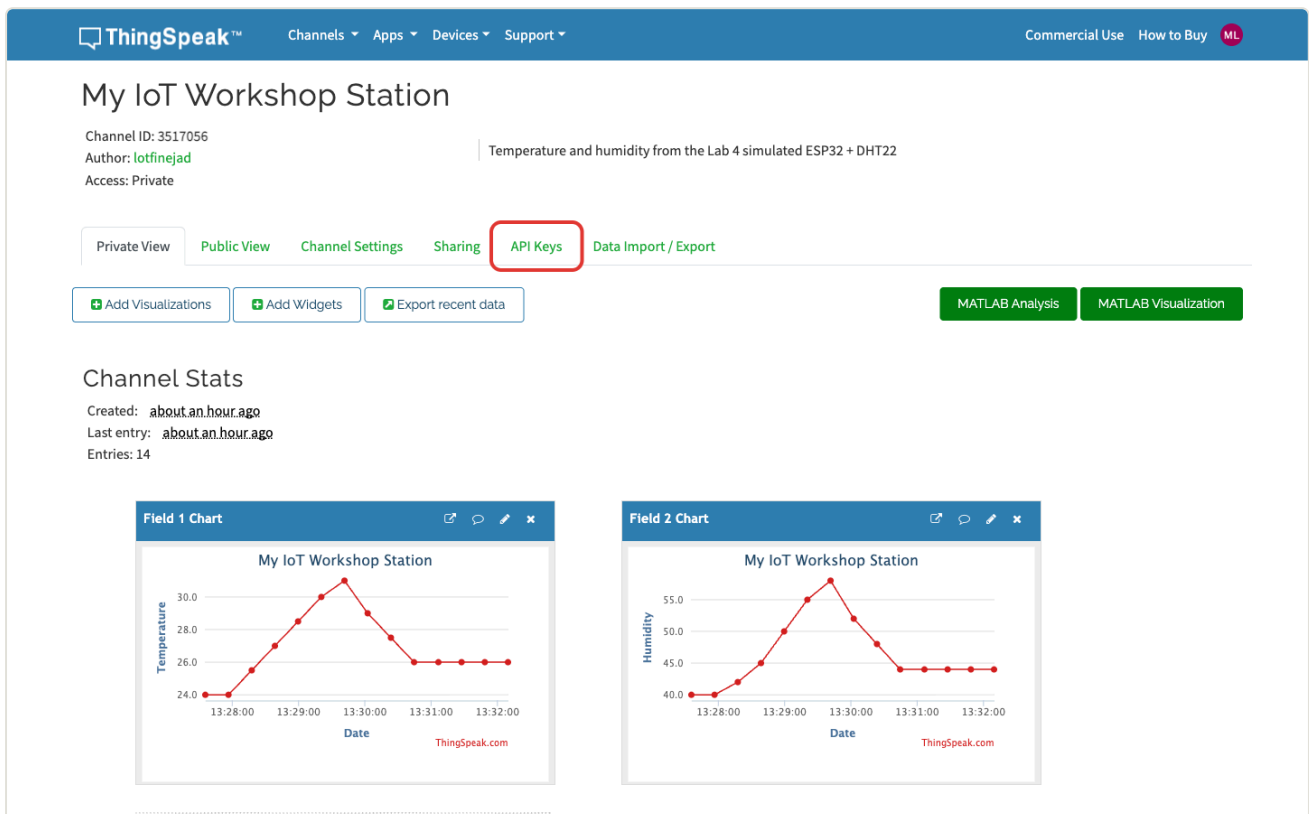
See [Get Started with ThingSpeak™](#) for an example of measuring dew point from a weather station that acquires data from an Arduino® device.

[Learn More](#)

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6. Your new channel's page opens, with a chart for each field. The charts stay empty until your device sends data in Part B (the screenshot was taken afterwards, so its charts already have points). Click the **API Keys** tab.



7. Copy the **Write API Key**. You'll need it in the next part.

ThingSpeak™

Channels Apps Devices Support

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My IoT Workshop Station

Channel ID: 3517056

Author: lotfinejad

Access: Private

Temperature and humidity from the Lab 4 simulated ESP32 + DHT22

Private View

Public View

Channel Settings

Sharing

API Keys

Data Import / Export

Write API Key

Key

XXXXXXXXXXXXXXXXXX

Generate New Write API Key

Read API Keys

Key

XXXXXXXXXXXXXXXXXX

Note

Save Note

Delete API Key

Add New Read API Key

Help

API keys enable you to write data to a channel or read data from a private channel. API keys are auto-generated when you create a new channel.

API Keys Settings

- Write API Key: Use this key to write data to a channel. If you feel your key has been compromised, click Generate New Write API Key.
- Read API Keys: Use this key to allow other people to view your private channel feeds and charts. Click Generate New Read API Key to generate an additional read key for the channel.
- Note: Use this field to enter information about channel read keys. For example, add notes to keep track of users with access to your channel.

API Requests

Write a Channel Feed

GET https://api.thingspeak.com/update?api_key=XXXXXXXXXXXXXXXXXX&field1=0

Read a Channel Feed

GET https://api.thingspeak.com/channels/3517056/feeds.json?api_key=XXXXXXXXXX

Read a Channel Field

GET https://api.thingspeak.com/channels/3517056/fields/1.json?api_key=XXXX

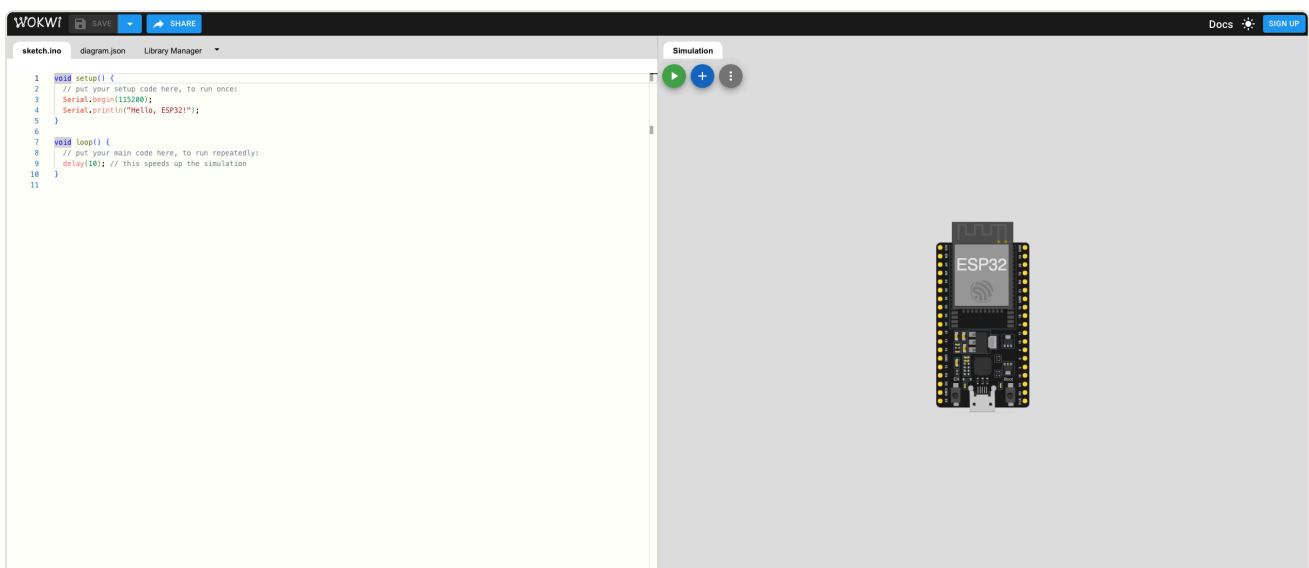
Read Channel Status Updates

GET https://api.thingspeak.com/channels/3517056/status.json?api_key=XXXX

Treat this key like a password: anyone who has it can write data to your channel. (The keys are hidden in the screenshot for that reason.)

Part B — Send data from the simulator (20 min)

1. Open a new ESP32 project

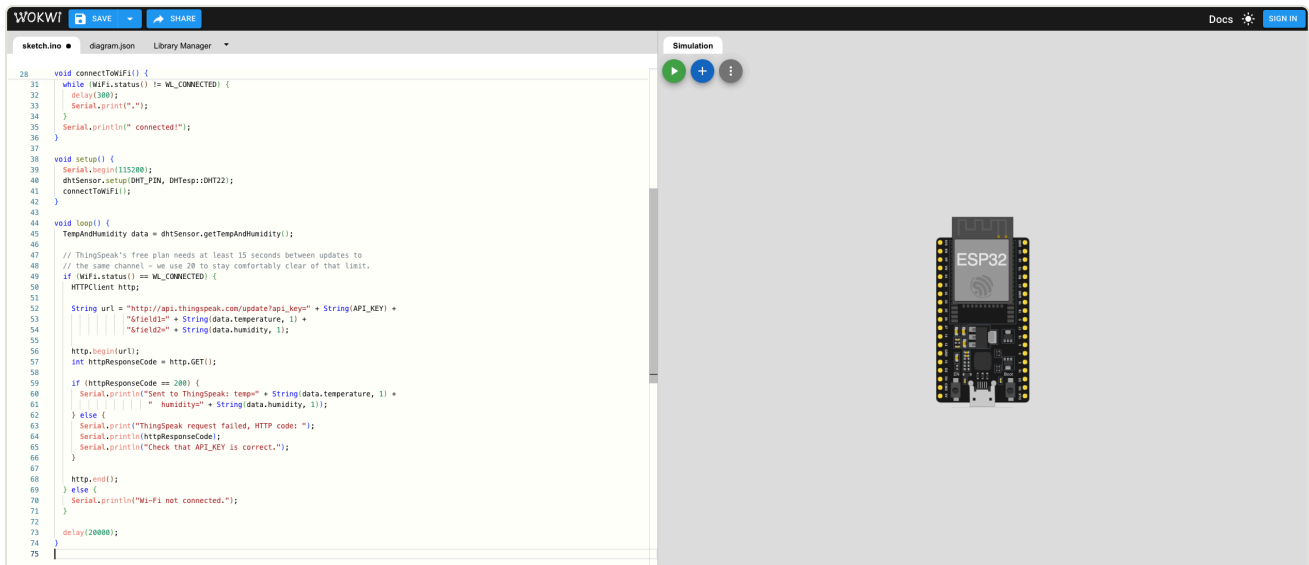


2. Paste the code — with your API key

Copy `sketch.ino` into Wokwi's **sketch.ino** tab. Before pressing Play, find this line near the top:

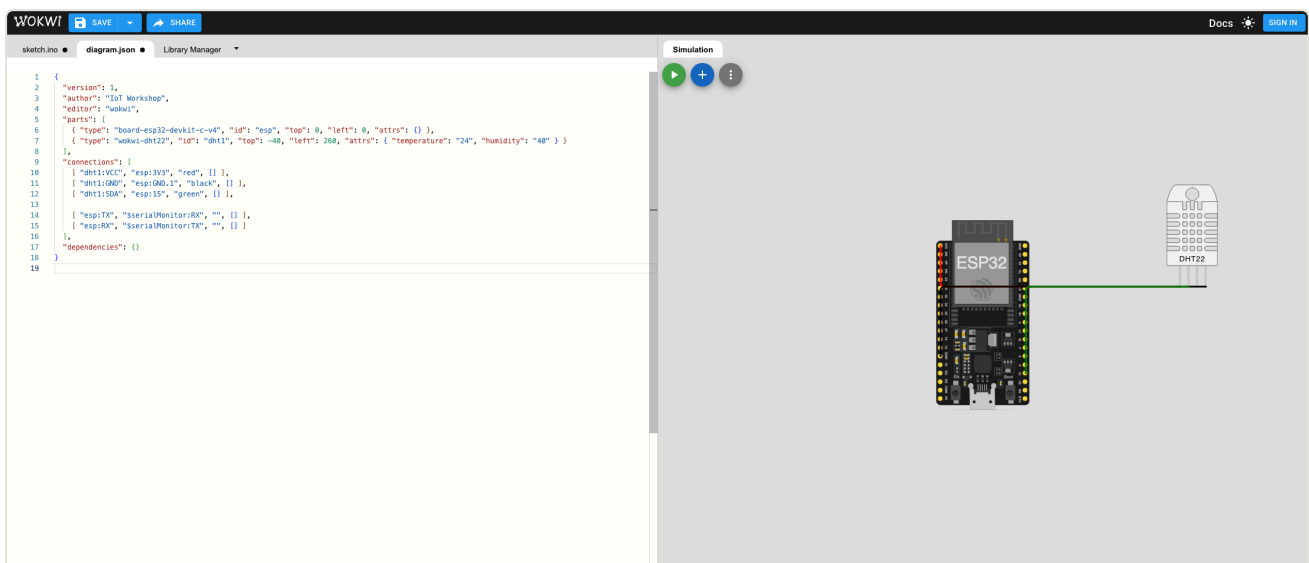
```
const char* API_KEY = "YOUR_WRITE_API_KEY";
```

Replace `YOUR_WRITE_API_KEY` with the key you copied in Part A (keep the quotes).



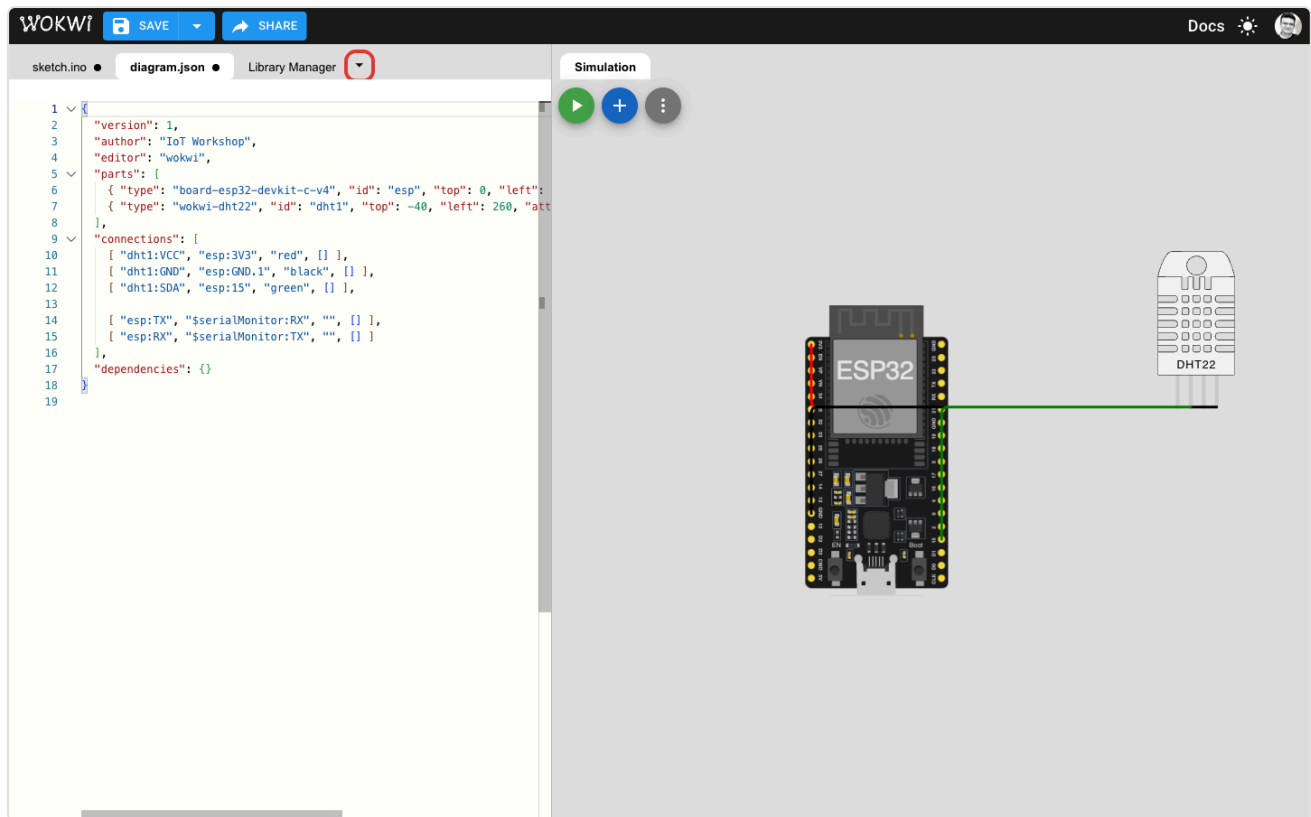
3. Paste the circuit

Copy `diagram.json` into Wokwi's `diagram.json` tab.

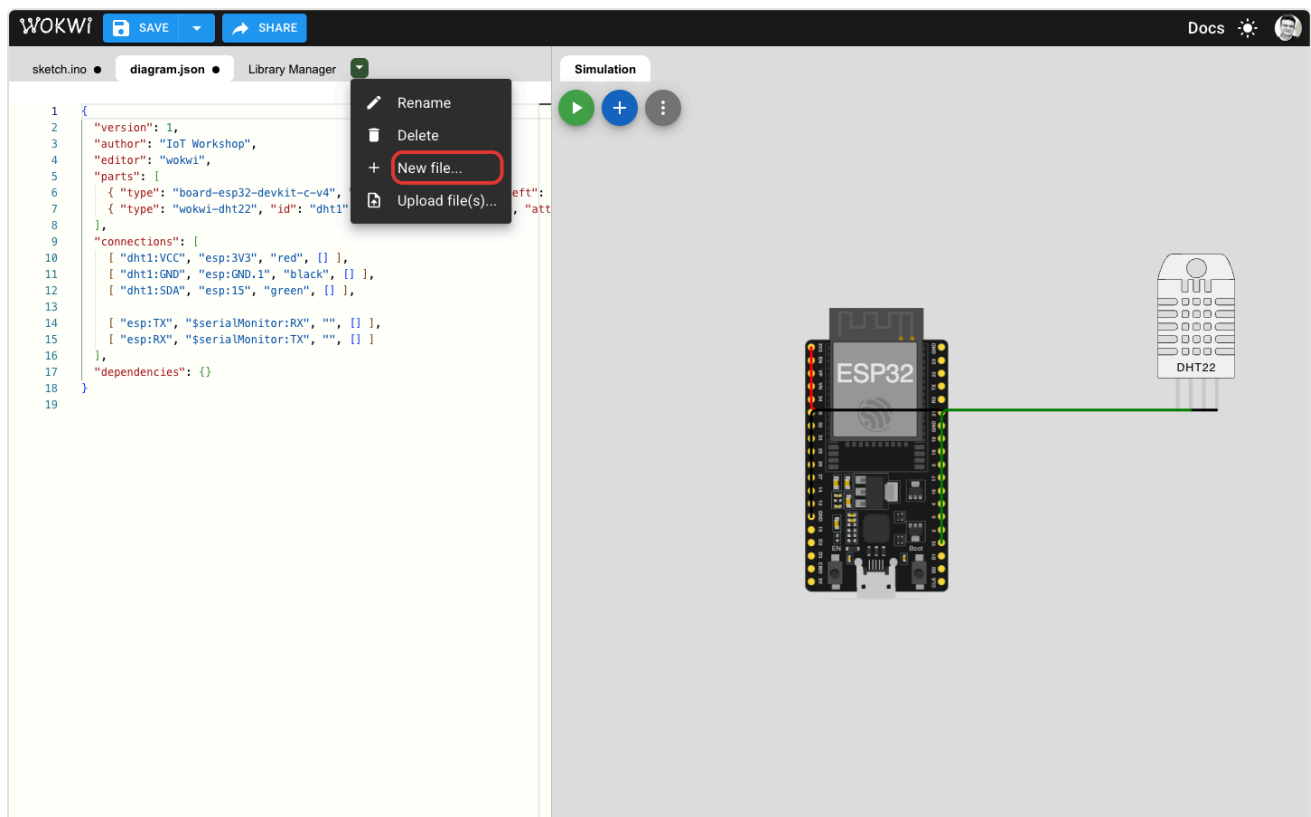


4. Add the library list

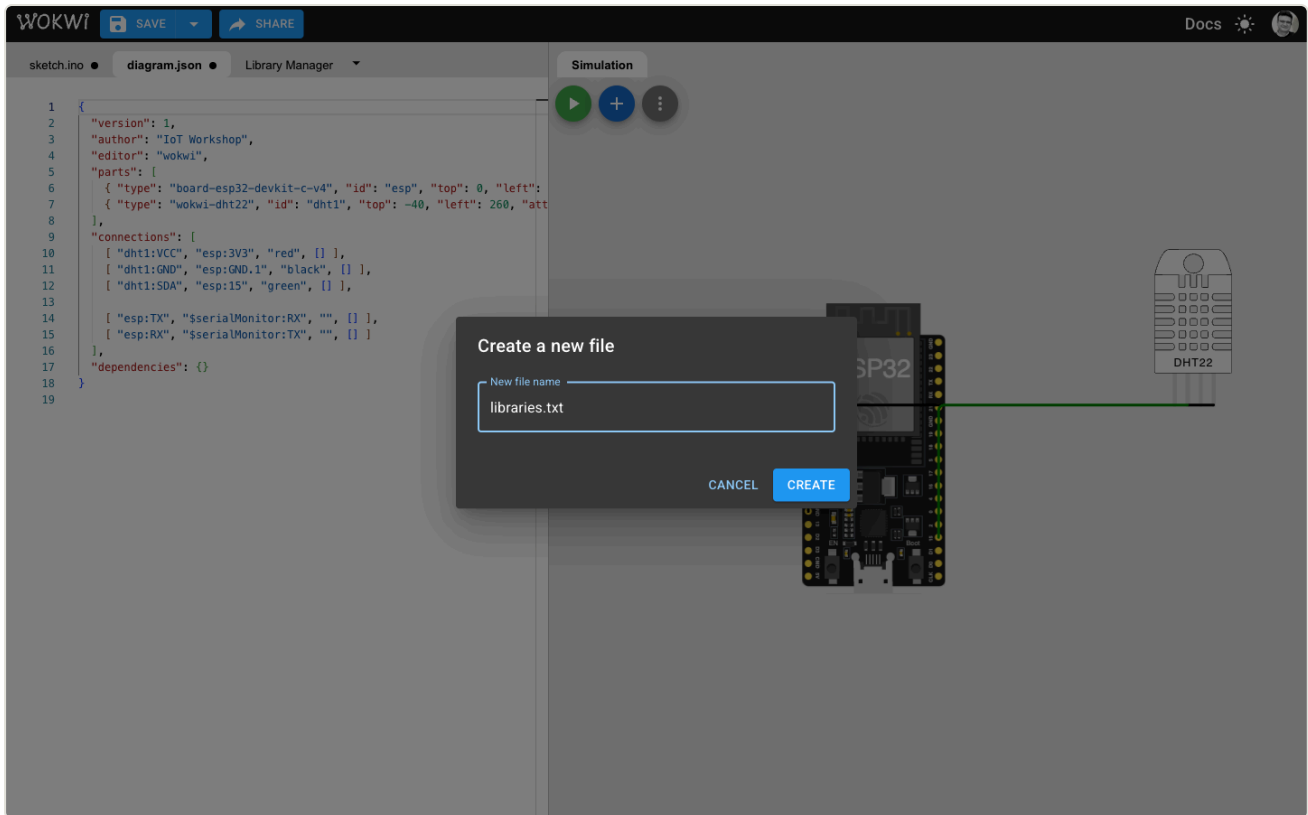
1. Click the small ▼ next to the **Library Manager** tab.



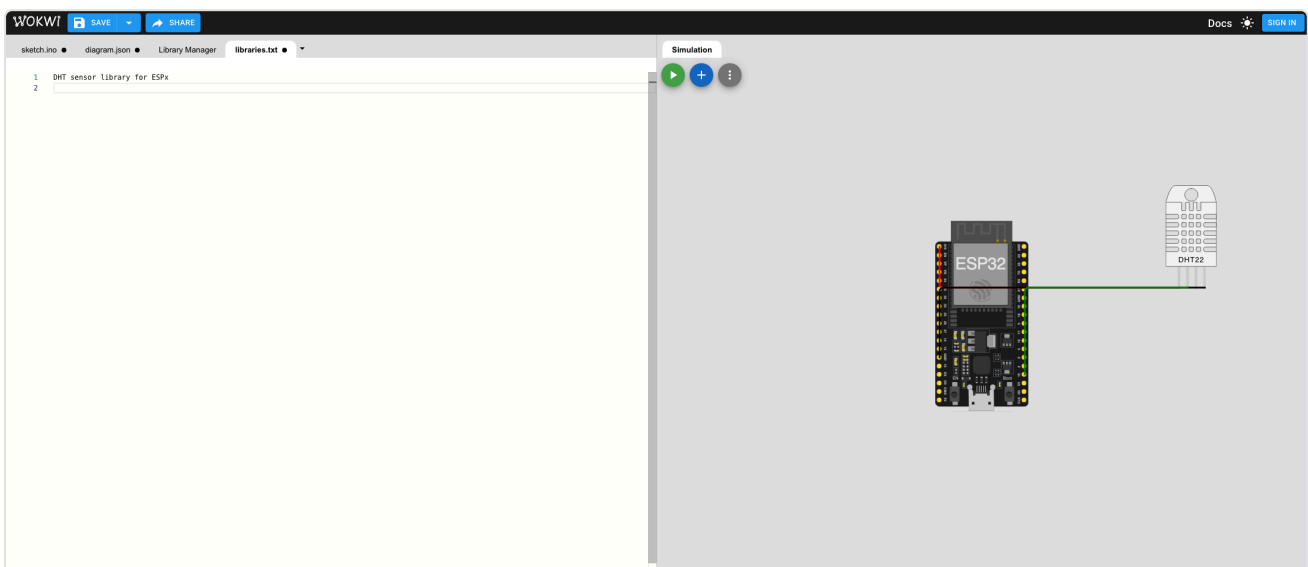
2. Choose New file....



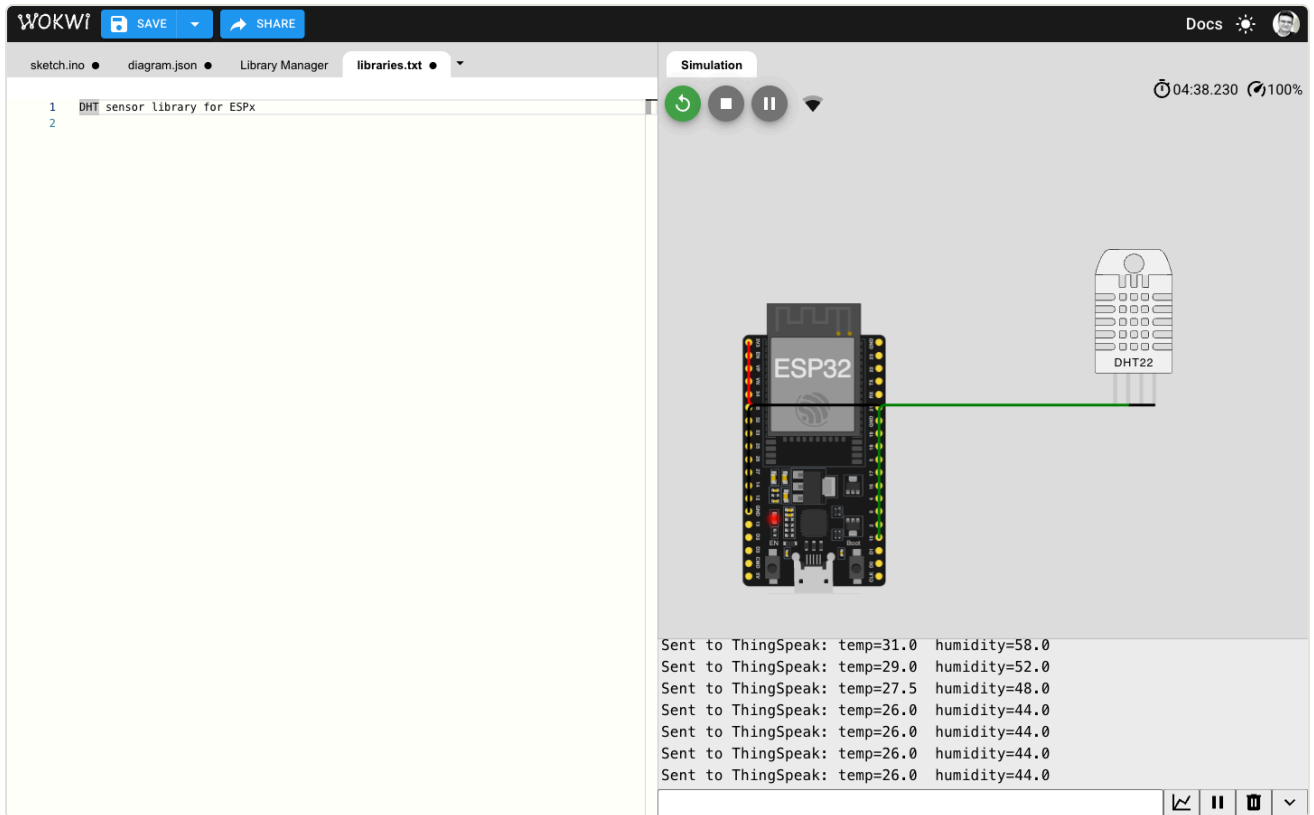
3. Type the name exactly `libraries.txt` and click **CREATE** (or press Enter).



4. Copy the contents of `libraries.txt` from this folder and paste them into the new `libraries.txt` tab.

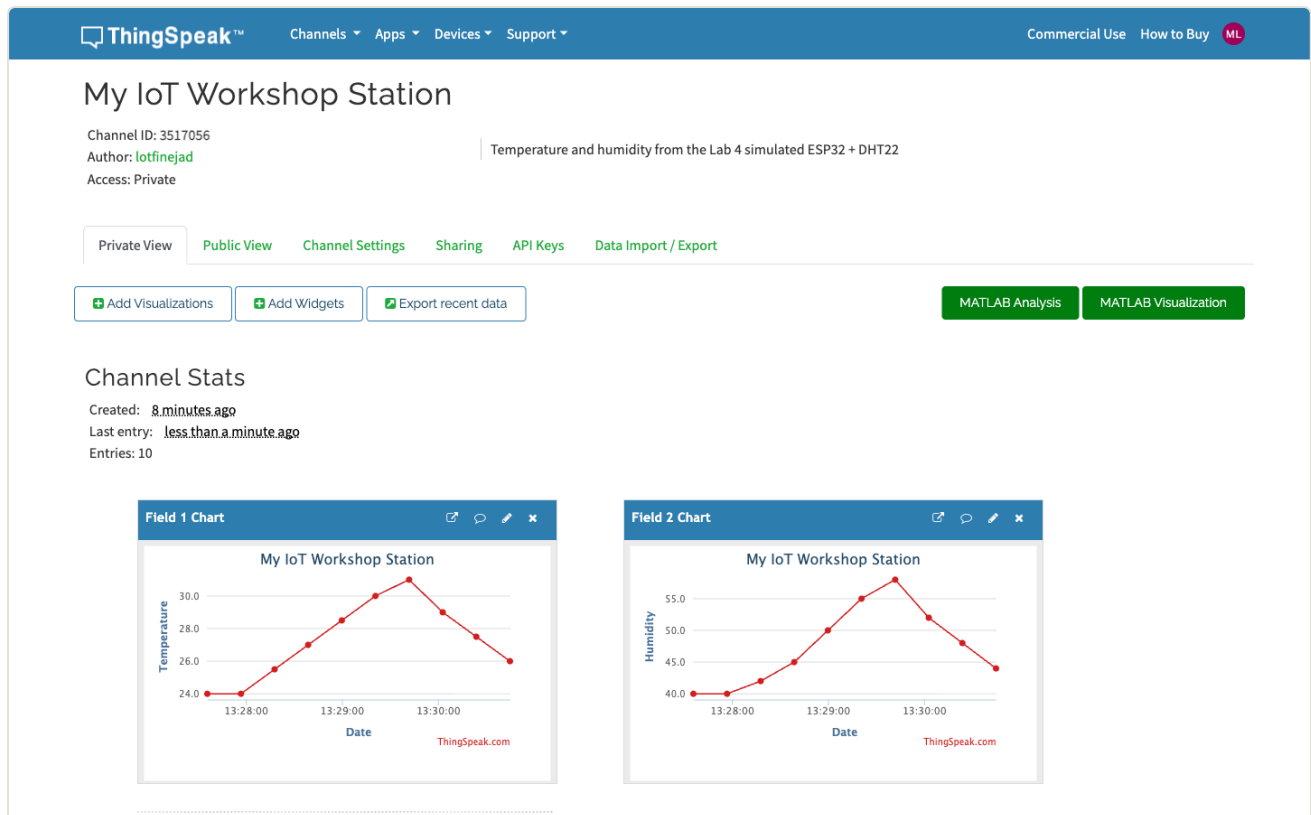


5. Press Play



The Serial Monitor should print `connected!` for Wi-Fi, then start printing `Sent to ThingSpeak: ...` every 20 seconds — or a clear error if the API key is wrong (see Troubleshooting).

6. Go back to your ThingSpeak channel page (**Channels** → **My Channels** → your channel) and watch **Field 1 Chart** and **Field 2 Chart** — new points should appear as your simulated device sends data. You may need to click the small refresh icon on each chart.



In this example, the DHT22's temperature and humidity sliders were moved between updates (see **Try it** below), so the charts rise and fall. If you don't touch the sliders, you'll see a flat line at 24 °C and 40% — that's correct too.

What you should see

With a correct API key:

```
Connecting to Wi-Fi..... connected!  
Sent to ThingSpeak: temp=24.0 humidity=40.0  
Sent to ThingSpeak: temp=24.0 humidity=40.0
```

With the placeholder key still in place (i.e. if you skip step 2), you'll correctly see an error instead — that's the code's error-handling working as designed, not a bug:

```
Connecting to Wi-Fi..... connected!  
ThingSpeak request failed, HTTP code: 400  
Check that API_KEY is correct.
```

Try it

- Click the DHT22 sensor in the Wokwi diagram and drag the temperature slider up and down. Wait 20 seconds and watch your ThingSpeak graph follow.

Understand the code

- `HTTPClient` is the ESP32's built-in tool for making web requests, the same kind your browser makes when you visit a page.
- ThingSpeak's API is deliberately simple: one web address (URL) with the values built right into it — `.../update?api_key=...&field1=...&field2=...`. This style of API, where you fetch a URL to do an action, is what Module 4 called **REST**.
- `http.GET()` returns a numeric response code. `200` means success — this is the same status code your browser gets for a page that loaded fine. If you see a different number, the most common cause is a wrong API key.
- The 20-second delay isn't arbitrary: ThingSpeak's free plan rejects updates sent less than 15 seconds apart on the same channel.

Troubleshooting

- **Build failed! ... DHTesp.h: No such file or directory**. You skipped step 4, or `libraries.txt` doesn't contain exactly `DHT sensor library for ESPx`.
- **ThingSpeak request failed, HTTP code: 400 (or similar)**. Your `API_KEY` is missing, wrong, or still the placeholder — go back to Part A step 7 and copy the **Write** key (not the Read key) exactly.
- **Chart doesn't update**. ThingSpeak's free plan needs 15+ seconds between updates on the same channel — the code already waits 20, so give it a full cycle, and try the small refresh icon on the chart itself.

Challenge (if you finish early)

- Add a third field to your channel (e.g. "Heat Index" or just a copy of temperature in Fahrenheit) and send it as `field3` in the URL.
- Look at your channel's **API** tab → **Read** — try fetching your own channel's latest data back with a browser, outside of the code.